

Probabilistic Forecasting of S&P 500 Stocks Using Conditioned Normalizing Flows

Miles Yamner, Stephen Slade (advisor)

February 2025

1 Background

Financial forecasting plays a key role in decision-making for investors, risk management, and algorithmic trading. Moreover, the decisions made by large investment firms have a substantial impact on the market and economy, underlying the importance of accurate and explainable forecasting.

1.1 Issues with Conventional Time Series Modeling

Traditional time series forecasting methods typically provide univariate point forecasts. Such a method usually necessitates training an individual model per time series, leading to low scalability. Additionally, many classical models still used today assume independence between assets, failing to capture sector-wide market movements and macro factors.

1.2 Traditional Financial Forecasting Methods and Their Shortcomings

Traditional statistical forecasting models, such as ARIMA, assume simple relationships in time-series data and fail to capture complex dependencies within datasets. ARIMA assumes linear relationships in time series data; however, as many financial factors are inherently nonlinear, ARIMA tends to fall short during non-stationary periods of economic development.

Deep learning models such as LSTM networks and Transformers enhance predictive accuracy by capturing more complex dependencies and long-range relationships in our data. However, these models typically rely on predefined probability distributions (e.g., Gaussian) for generating forecasts, restricting their ability to fully model the true, distribution of financial returns.

1.3 Normalizing Flows in Time-Series Forecasting

Normalizing Flows (NFs) provide a flexible framework for modeling complex probability distributions in time-series forecasting by learning a sequence of invertible transformations. Given a time series observation X_t drawn from an unknown distribution $p_X(X)$, NFs define a bijective mapping f that transforms

X_t into a latent variable Z_t with a simpler, known base distribution (typically a standard Gaussian $\mathcal{N}(0, I)$):

$$Z_t = f(X_t), \quad X_t = f^{-1}(Z_t)$$

This allows the original data distribution to be recovered exactly. Using the change of variables formula, the probability density function (PDF) of X_t is then given by:

$$p_X(X_t) = p_Z(Z_t) \left| \det \frac{df(X_t)}{dX_t} \right|^{-1}$$

where $\det \frac{df(X_t)}{dX_t}$ is the determinant of the Jacobian of the transformation f . This property enables exact likelihood computation, making NF-based models well-suited for density estimation and probabilistic forecasting. However, note that NFs alone still suffer from dependency issues, as they do not explicitly model sequential dependencies in time-series data or external market influences.

1.4 Conditioned Normalizing Flows for Time-Series Forecasting

Conditioned Normalizing Flows (CNFs) extend NFs by incorporating external conditioning variables into the transformation function, allowing the learned probability distribution to adapt based on context. While standard NFs assume that each observation X_t follows a static transformation function, CNFs introduce a dependency on a conditioning input C_t :

$$Z_t = f(X_t|C_t)$$

where C_t represents structured external information, such as Historical Lagged Features or Macroeconomic indicators (e.g., interest rates, VIX, LIBOR) that influence financial trends. This addresses several key limitations of both traditional forecasting models and standard NFs:

1. **Sequential Dependency Modeling:** CNFs explicitly condition future distributions on past values, capturing time-dependency, rather than treating each X_t independently.
2. **Cross-Asset Dependency Handling:** Sector-wide influences and stock correlations are accounted for by including external covariates.
3. **Market Regime Adaptability:** This is particularly useful in non stationary markets. By conditioning on macroeconomic indicators, CNFs should provide more stable forecasts that adapt to changing regimes.

2 Project Description

This project aims to explore the application of CNFs to model the joint probability distribution of S&P 500 stocks, enabling probabilistic forecasting along with

an uncertainty estimation. The implementation will leverage the PyTorch-based Probabilistic Time Series forecasting framework (PyTorchTS) based on GluonTS backend¹, which provides efficient tools for time-series modeling, dataset transformation, and back-testing.

The dataset will consist of historical price data for all 500 S&P 500 stocks, along with trading volume, sector classification, and macroeconomic indicators such as interest rates, VIX, and inflation data sourced from Yahoo Finance, Quandl, and other comparable data platforms. Data preprocessing will include computing log returns, sector-based embeddings, volatility metrics, and possibly others, ensuring that our model effectively captures market dependencies and risk factors.

To evaluate the effectiveness of CNFs for financial forecasting, the project will compare its performance against two baseline models: DeepVAR (A multivariate probabilistic forecasting model that captures dependencies across multiple time series), and a simple feed-forward neural network.

Performance evaluation will be based on probabilistic forecasting metrics such as Continuous Ranked Probability Score (CRPS) and log-likelihood estimation, as well as standard regression accuracy measures.

3 Proposed Schedule and Timeline

Week of	Task
Feb 2, 9	Literature review, data collection and preprocessing
Feb 16, 23	Modify TempFlow CNF model for S&P 500 data
Mar 2, 9	Train CNF model on S&P 500 data
Mar 16, 23	Evaluate CNF model performance and fine-tune parameters
Mar 30, Apr 6	Compare CNF model against baselines (DeepVAR, feed-forward)
Apr 13	Draft final report and present preliminary results to advisor
Apr 20	Revise report based on feedback, prepare poster presentation
Apr 27	Finalize submission and complete presentation materials

4 Deliverables

1. Research report detailing methodology, experimental setup, and results.
2. Forecasting visualizations with confidence intervals and cross-stock dependencies.
3. Comparative analysis of CNF performance against baseline models (DeepVAR, Simple Feedforward).
4. Final presentation and project poster summarizing findings.

¹Kashif Rasul, Calvin Seward, et al. *PyTorch-based Probabilistic Time Series Forecasting Framework*. GitHub, 2020. Available at: <https://github.com/zalandoresearch/pytorch-ts>

5 Note about Joint Math-CS Major

The mathematical aspect will be integrated through probabilistic modeling, normalizing flow transformations, and density estimation techniques. Specifically, the project will involve maximum likelihood estimation, change-of-variables theorem for probability transformations, and Jacobian determinant computations in the context of CNFs. The computer science portion will be the implementation and ML theory backing the project.

References

- [1] Kashif Rasul, Abdul-Saboor Sheikh, Ingmar Schuster, Urs Bergmann, and Roland Vollgraf. Multivariate probabilistic time series forecasting via conditioned normalizing flows. *International Conference on Learning Representations (ICLR)*, 2021.
- [2] Jin Liu. Navigating the financial landscape: The power and limitations of the ARIMA model. *Highlights in Science, Engineering and Technology (IFMPT)*, Volume 88:747-752, 2024.
- [3] Adèle Gouttes, Kashif Rasul, Mateusz Koren, Johannes Stephan, and Tofigh Naghibi. Probabilistic time series forecasting with implicit quantile networks. *Proceedings of the 38th International Conference on Machine Learning (ICML)*, PMLR 139, 2021.